



Team 3 - AI-Empowered Space-Air-Ground Integrated Networks (SAGIN)

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The Aerial Edge: Powering 6G via UAVs & HAPS

Advanced Trajectory,
Beamforming, and Edge
Computing in the Air Layer

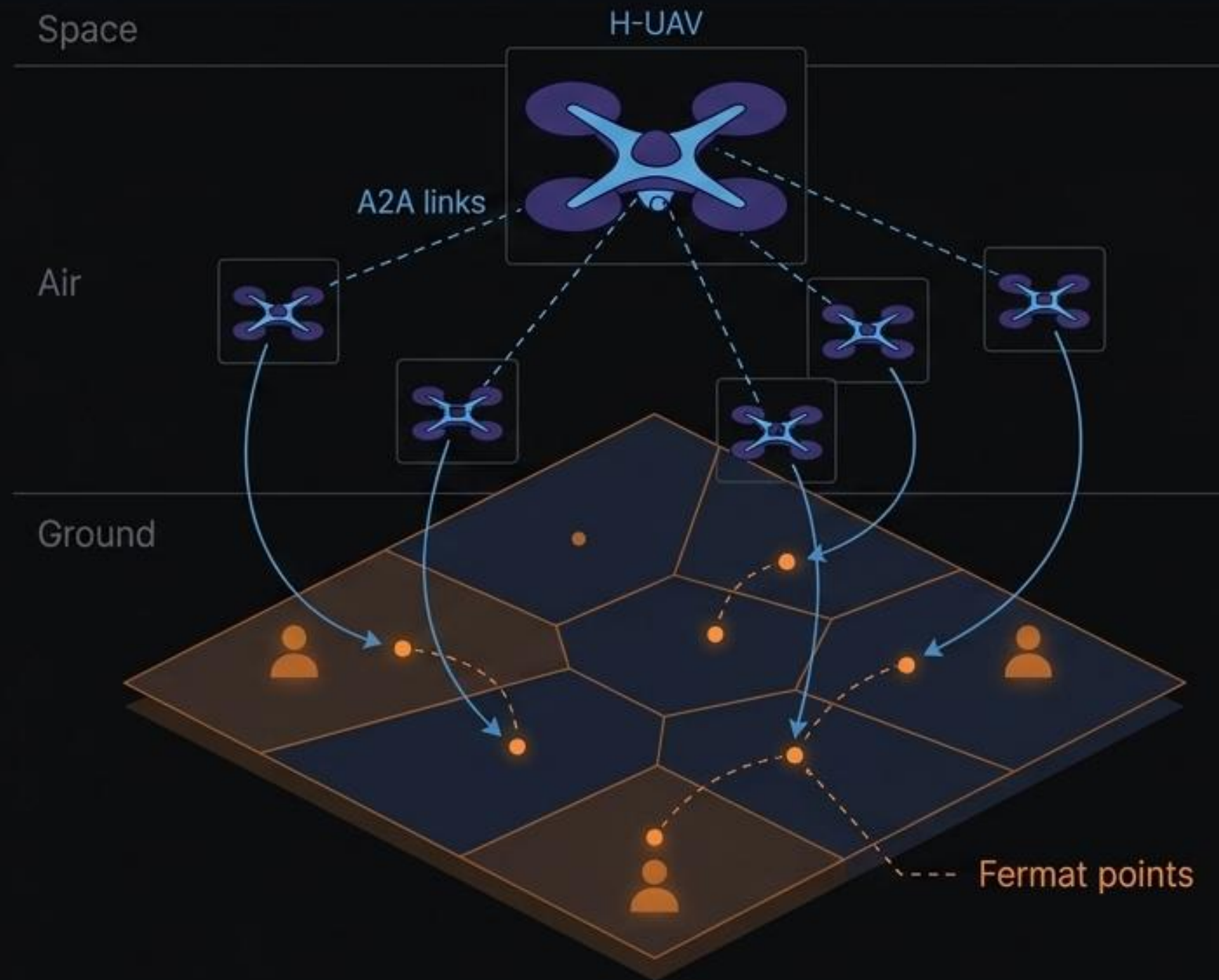


Optimizing UAV Swarm Trajectories for 6G Aerial Access

Balancing aerodynamic flight energy with data collection latency requires rigorous swarm intelligence.

Hierarchical Swarms: Head-UAVs (H-UAV) hover centrally while Terminal-UAVs (T-UAV) fly to optimal data collection points.

The Optimizer: The Improved Non-dominated Sorting Whale Optimization Algorithm (INS-WOA) computes Pareto-optimal flight paths.



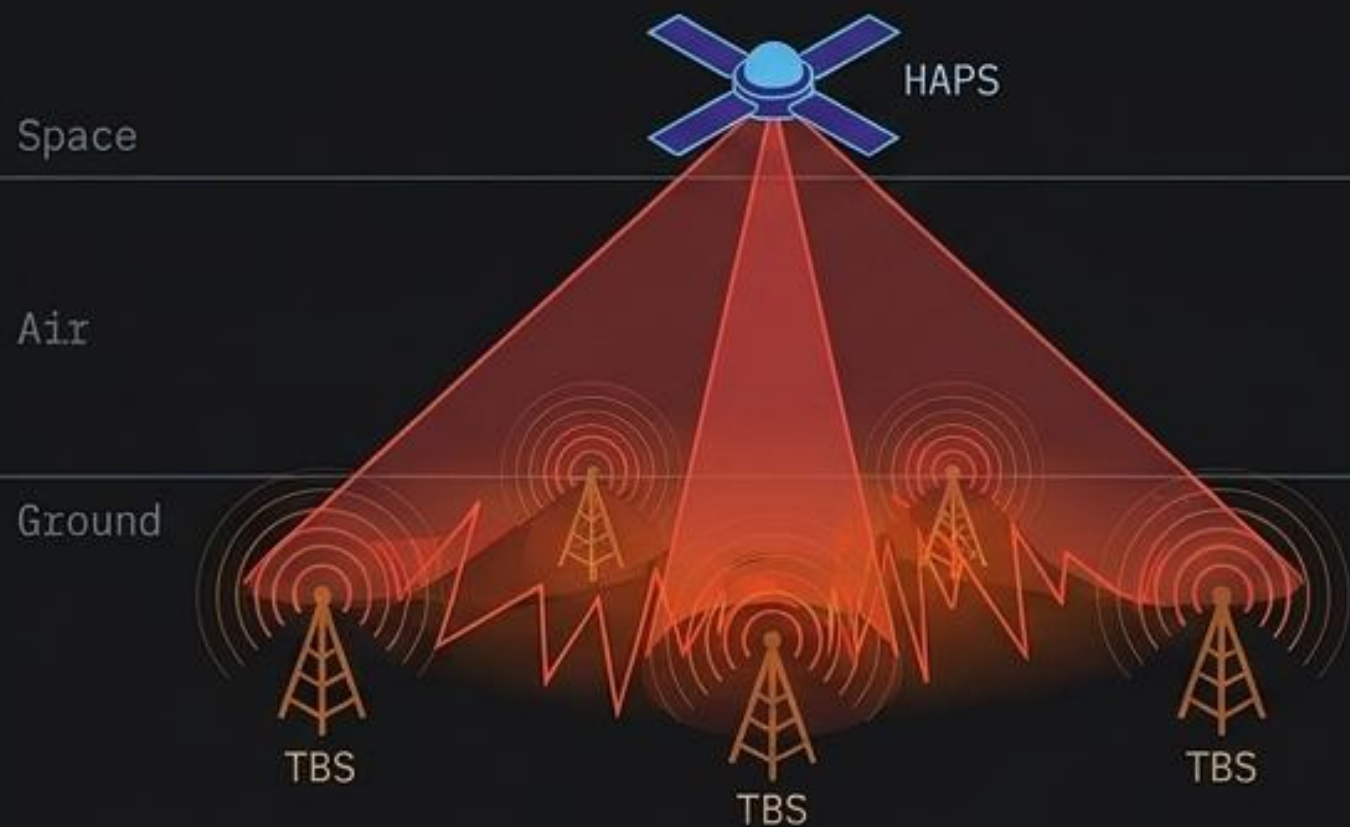
Managing Interference in High-Altitude Platform Stations (HAPS)

HAPS offer quasi-stationary positioning with lower latency than satellites, but generate severe inter-layer interference with terrestrial networks.

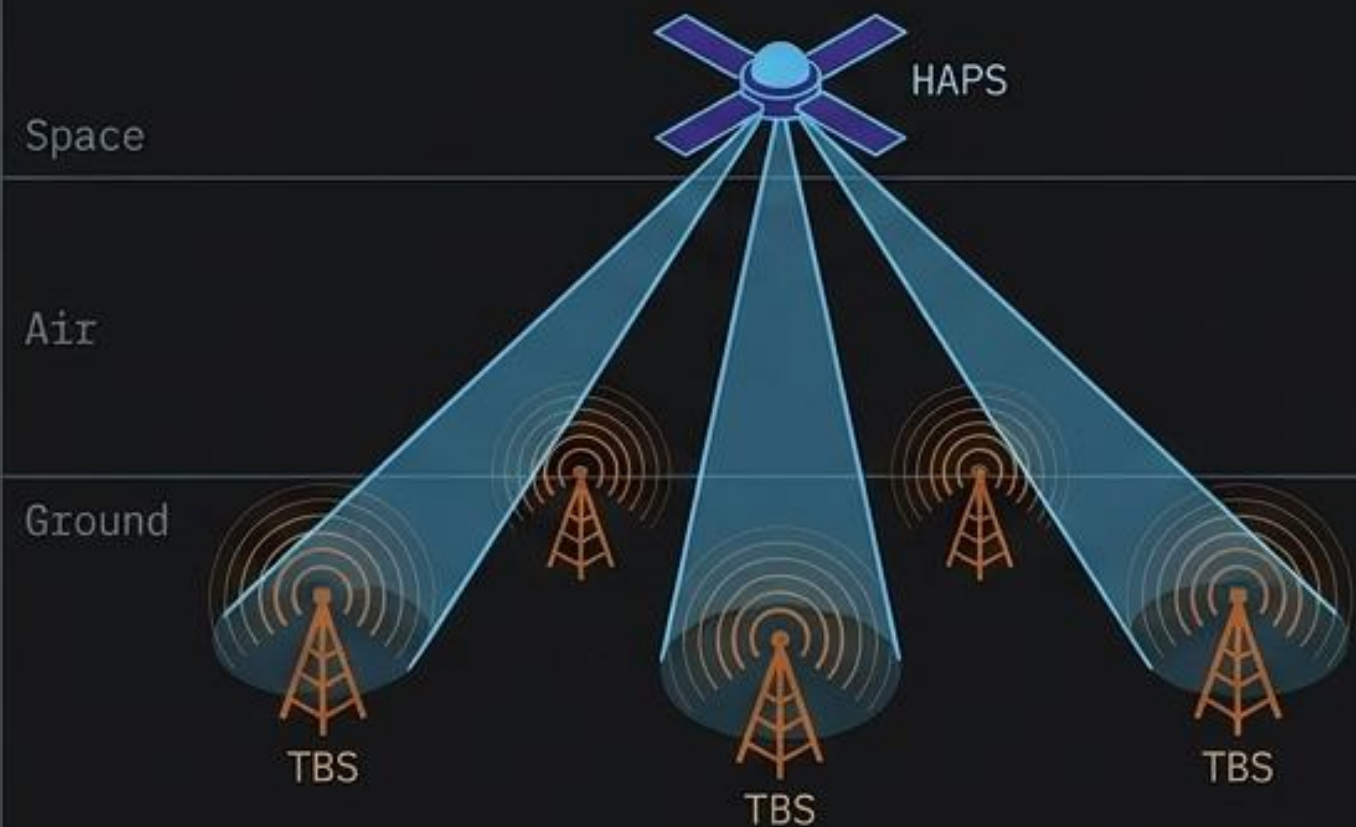
Dynamic Beamforming: Entropy-based Multi-Agent Deep Reinforcement Learning (DRL) adjusts complex beamforming vectors in real-time.

Result: Maximized Spectral Efficiency without disrupting ground-level Terrestrial Base Stations (TBS).

The Problem: Static Coverage

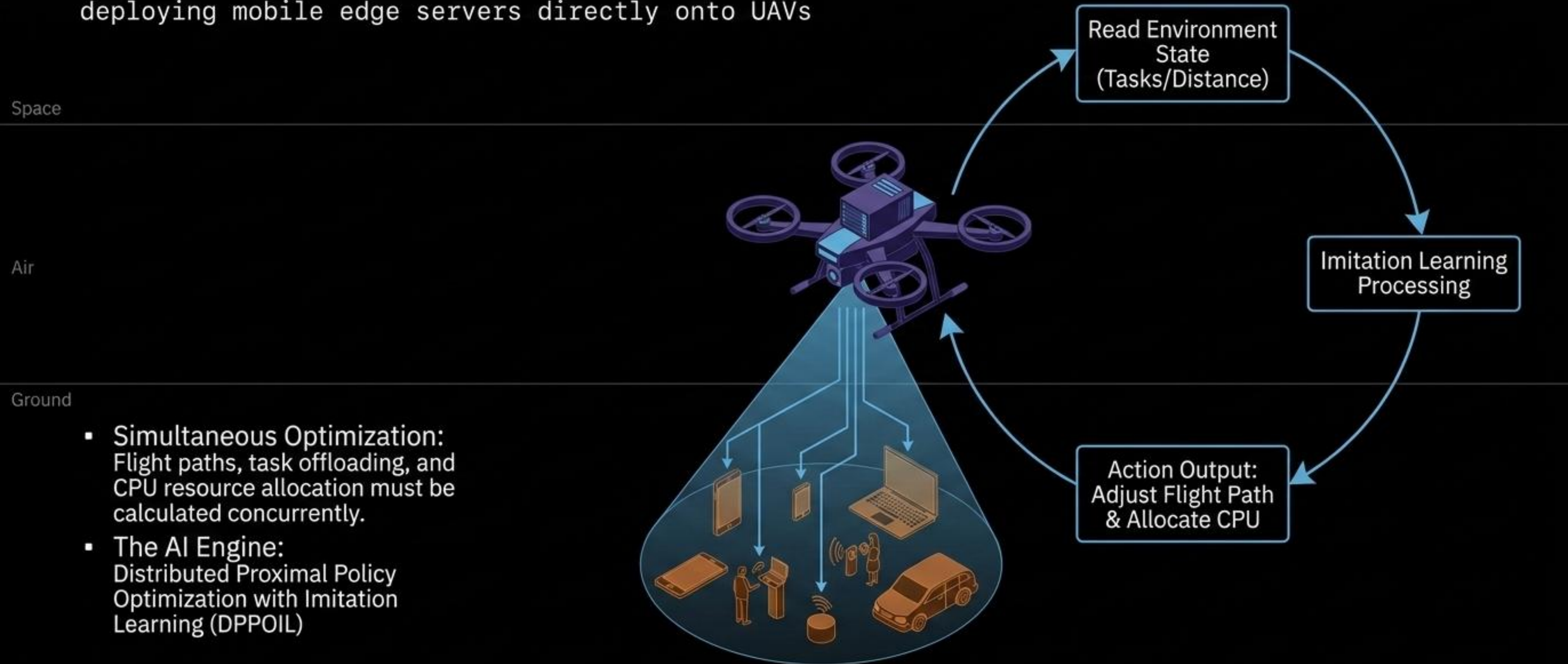


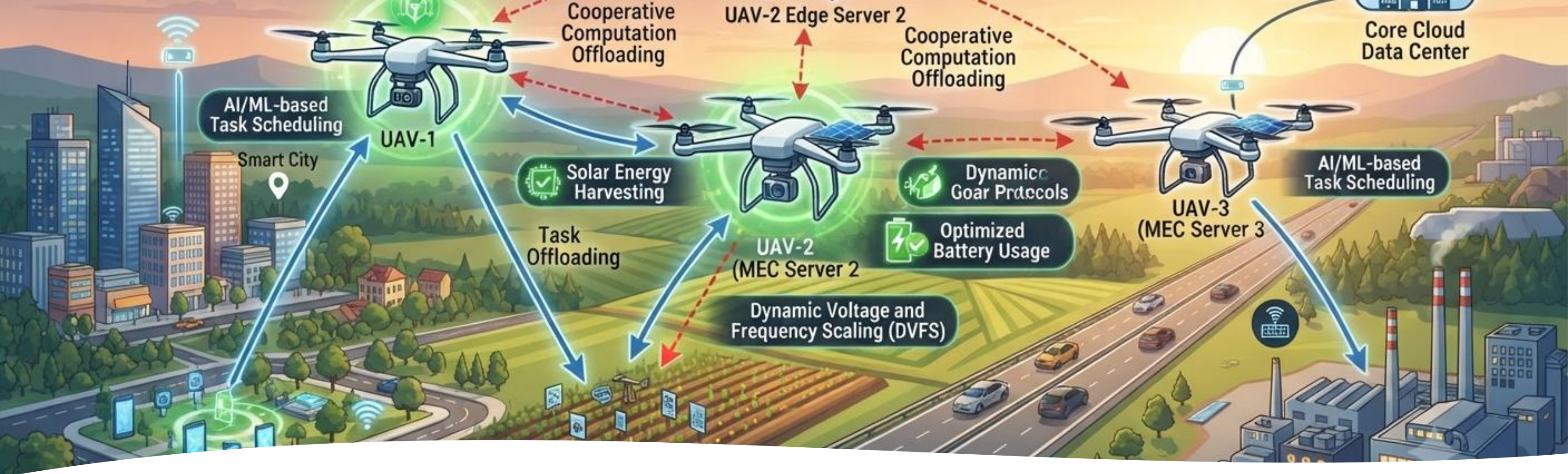
The Solution: Entropy DRL Beamforming



Bringing Computation to the Stratosphere

Bypassing the limits of fixed terrestrial MECs by deploying mobile edge servers directly onto UAVs





Joint Optimization in Multi-UAV MEC

- **Overcoming Siloed Approaches:** Traditional methods treat UAV flight paths and compute resource allocation separately, leading to systemic inefficiencies. Zhang et al. (2025) solve both challenges jointly using Deep Reinforcement Learning (DRL).
- **Real-Time Adaptive Control:** The intelligent model dynamically adapts UAV positioning and distributes computing resources to ground devices in real time, moving past rigid, fixed-rule allocations.
- **Measurable Performance Gains:** Achieves significantly more users served, lower overall energy consumption, and reduced communication delay compared to standard legacy optimization baselines.



- **Edge AI for Physical Layer Security in 6G Non-Terrestrial Networks**

Securing the Aerial Layer

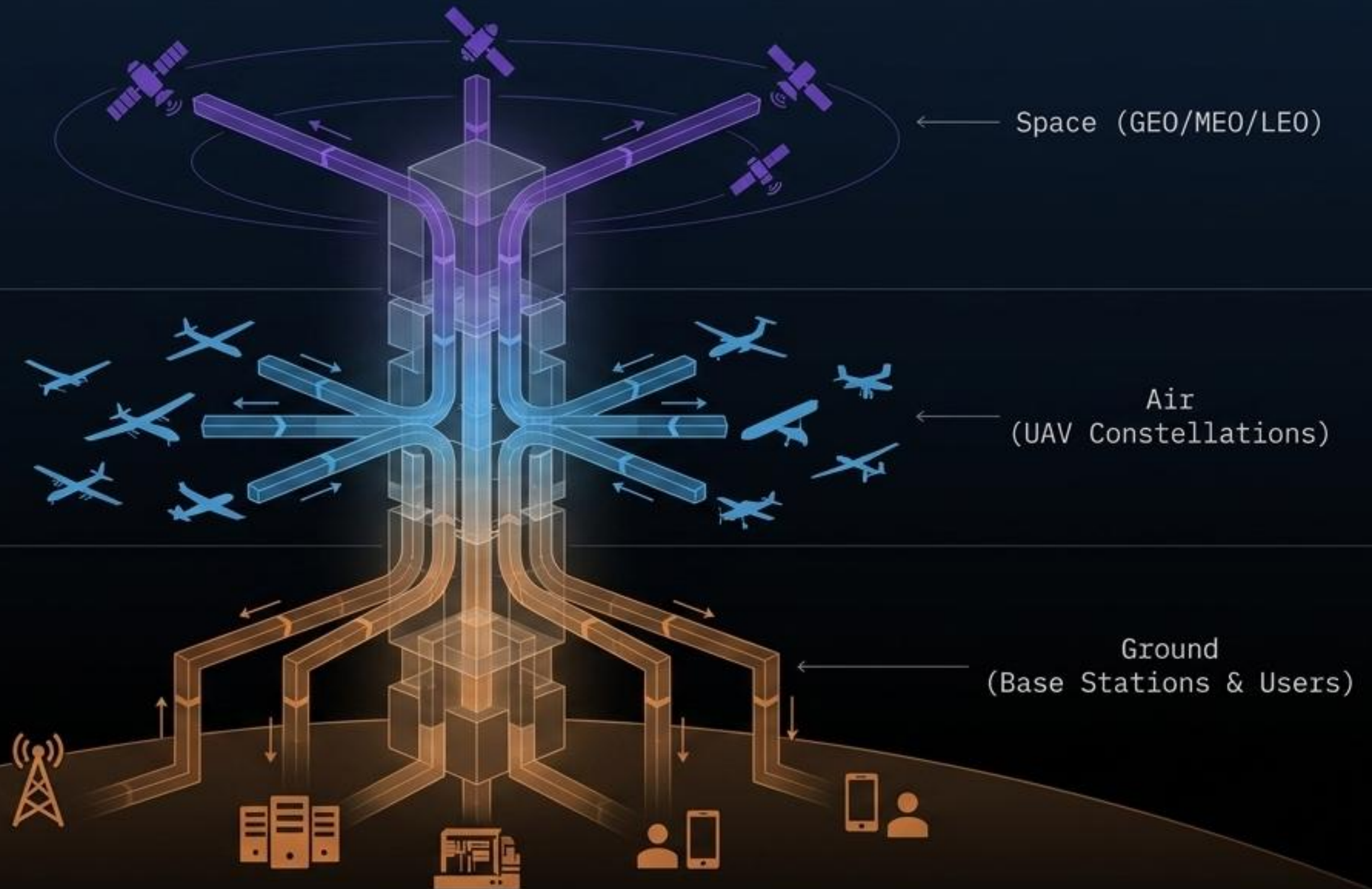
- **Expanding Threat Landscape:** As SAGIN connects satellites, drones, and HAPS, it becomes more exposed to interference, spoofing, jamming, and eavesdropping
- **Edge AI Defense:** Chou et al. (2024) propose using Edge AI to detect threats and secure communications closer to where data is generated instead of relying on a distant central system.
- **Privacy & Speed:** Techniques like federated learning and split deep learning keep processing local, improving both response speed and privacy

Algorithmic Approaches to Aerial Layer Challenges

Focus Area	Aerial Node	AI Model	Key Optimization Metric
Data Collection	UAV Swarm 	INS-WOA (Whale Optimization)	Flight Energy vs. Offloading Delay
Interference Management	High-Altitude Platform (HAPS) 	Multi-Agent DRL	Spectral Efficiency
Edge Computing	MEC-UAV 	DPPOIL (Imitation Learning)	Total Energy + Offloading Efficiency

The Integrated Brain: Orchestrating SAGIN

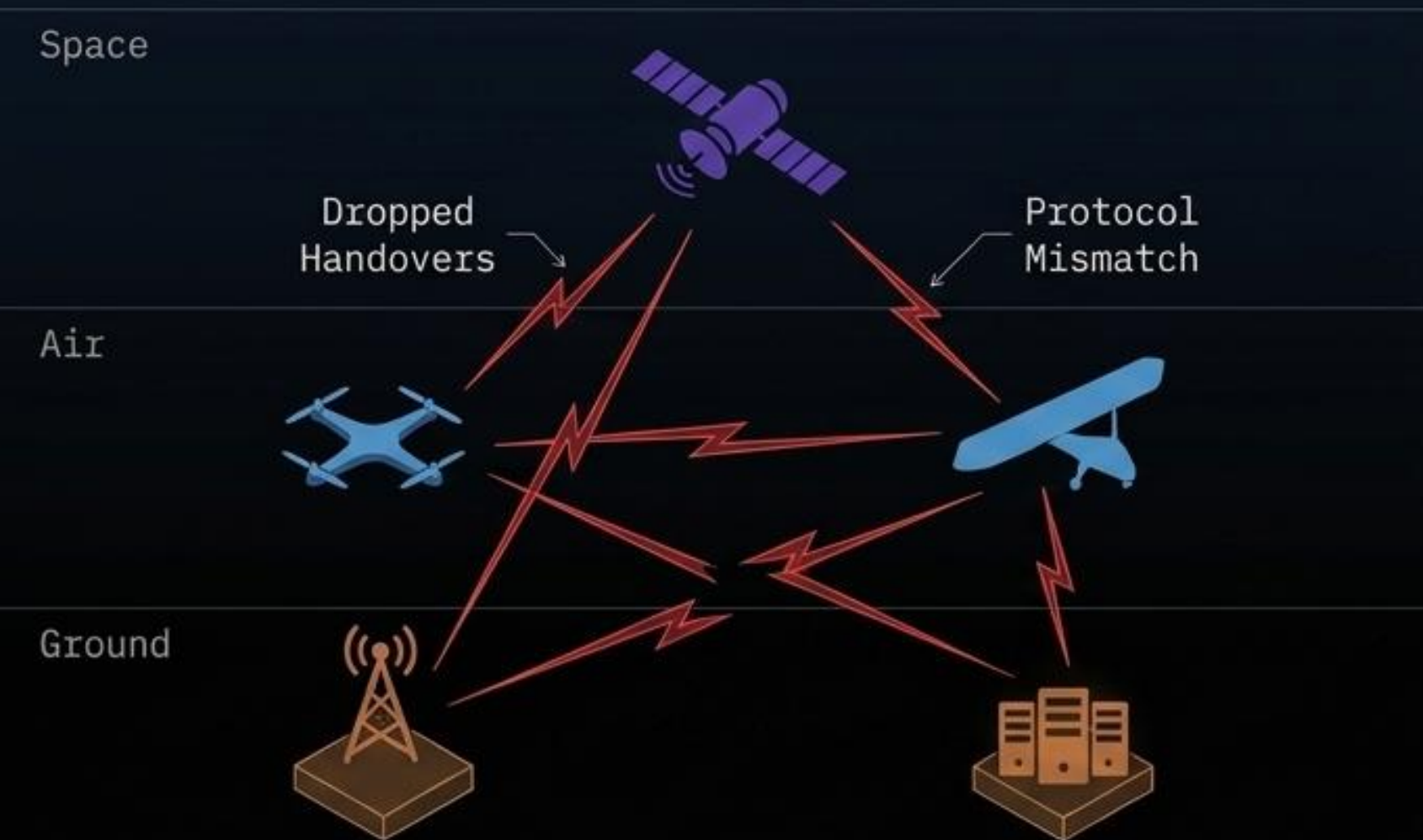
Managing Multi-Tier Complexity, Predictive Routing, and Privacy



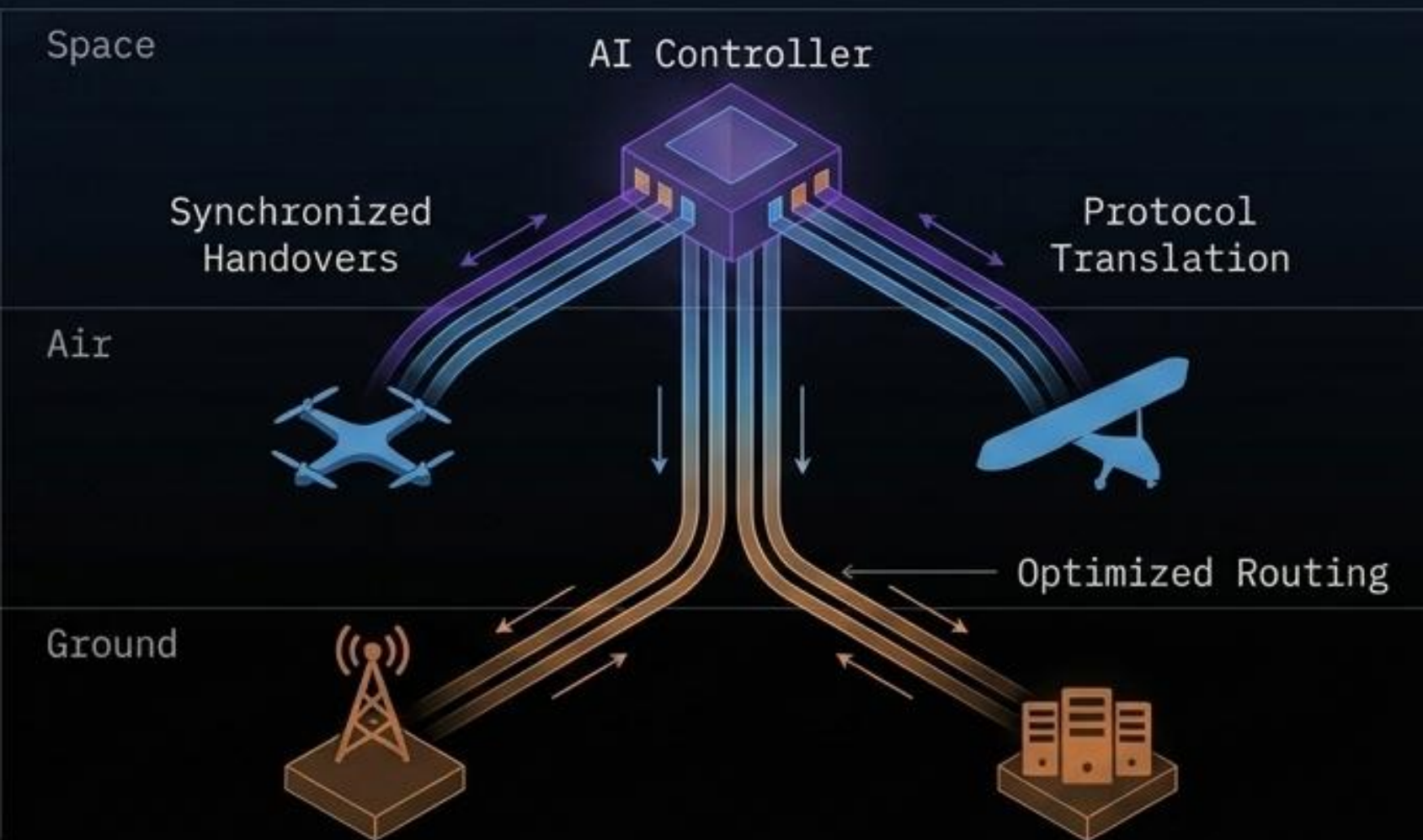
The Inherent Complexity of Multi-Tier Networks

- Heterogeneity: Diverse protocols across TCP/IP, satellite links, and aerial relays.
- Extreme Mobility: LEOs and UAVs are highly dynamic, requiring constant topological updates and handover management.
- The Control Dilemma: Distributed control reduces bottlenecks; centralized control simplifies global structure. SAGIN strictly requires both.

The Chaos



The Harmony



Hybrid Intelligence for Global Orchestration

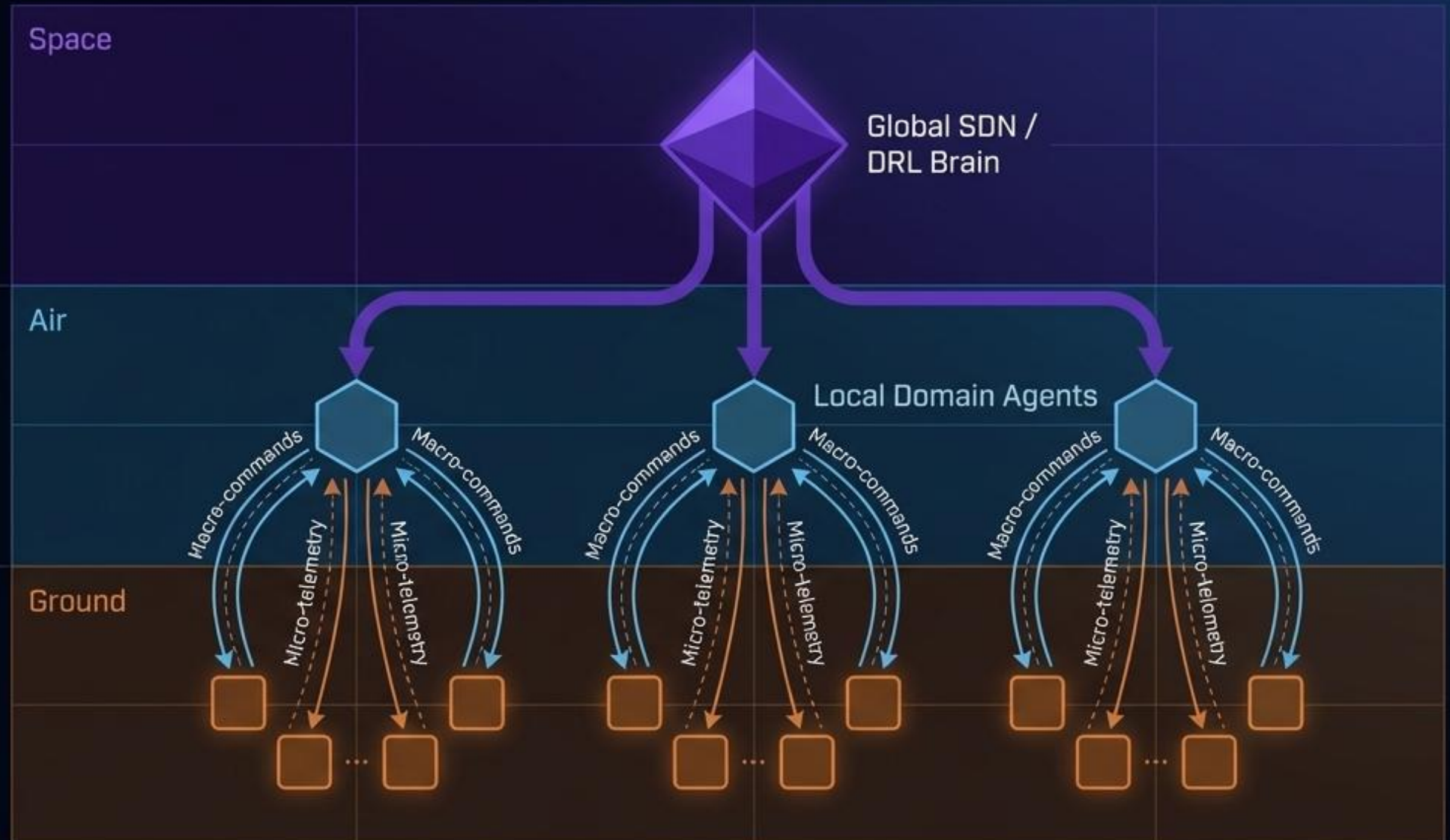
Hierarchical Hybrid Deep Reinforcement Learning (HHDRL) bridges the gap between distributed speed and centralized planning.

Global Agent (Centralized):

Macro-control, global SDN routing, and overarching resource allocation across the entire SAGIN.

Local Agents (Distributed):

Fast, fine-grained control within specific domains (e.g., specific UAV swarms or ground stations).



Orbital Spatiotemporal Awareness

Unlike ground networks, SAGIN topologies change predictably based on Kepler's laws of orbital mechanics.

Space

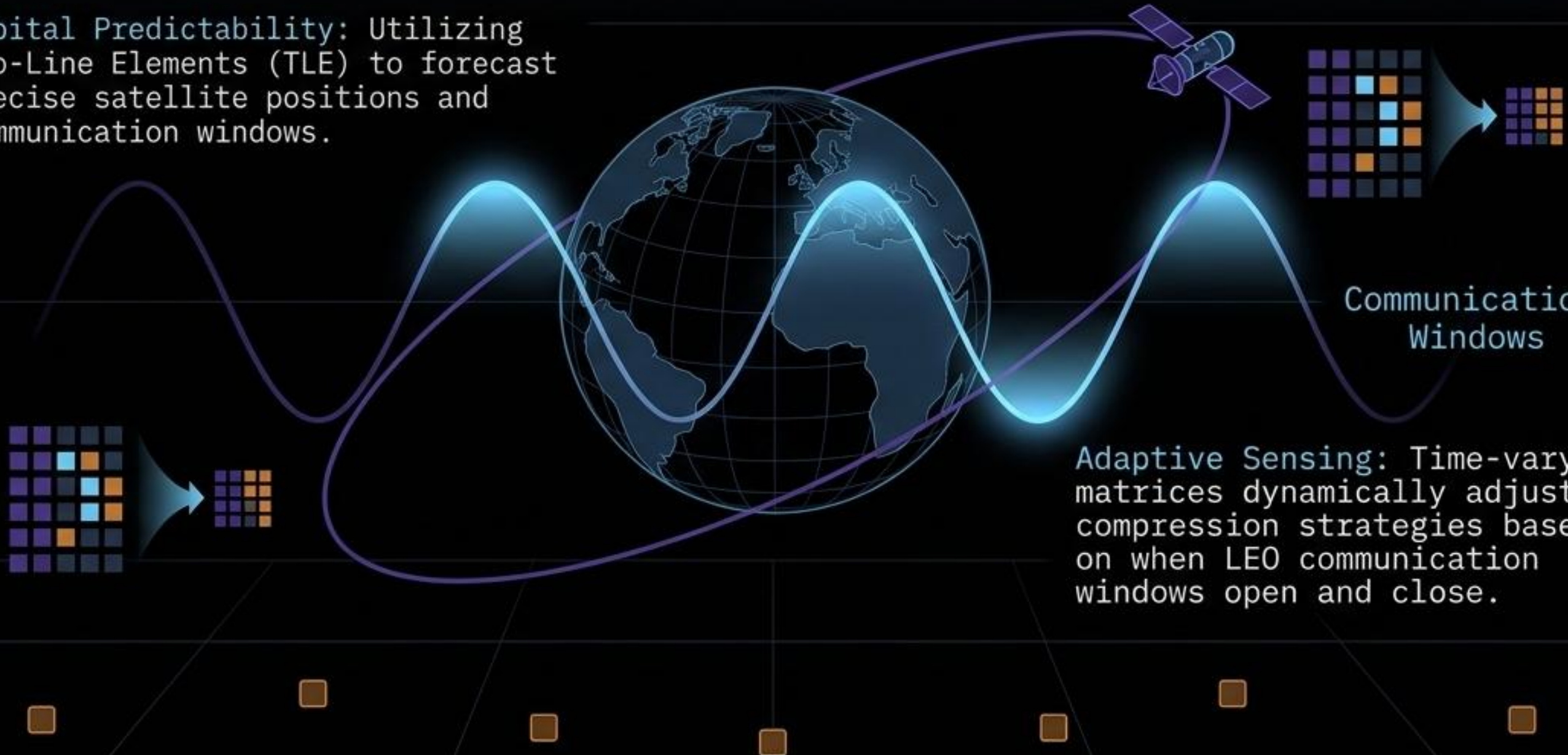
Orbital Predictability: Utilizing Two-Line Elements (TLE) to forecast precise satellite positions and communication windows.

Air

Communication Windows

Ground

Adaptive Sensing: Time-varying matrices dynamically adjust data compression strategies based on when LEO communication windows open and close.



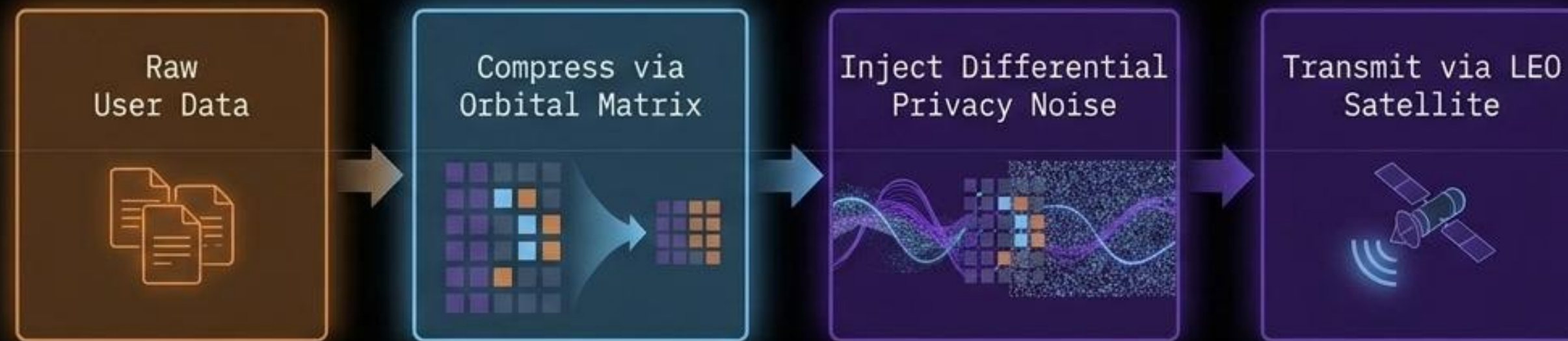
Privacy in the Stratosphere (ACDP-FL)

Federated Learning exposes model gradients to interception.
Adaptive Compressed Sensing Differential Privacy (ACDP-FL)
solves this by injecting noise within the compressed domain.

Space

Dynamic Balance: Achieves a 30-50% enhancement in communication efficiency while maintaining robust privacy.

Air



Privacy

Ground



The 6G Horizon: Quantum Scheduling & Cybertwins

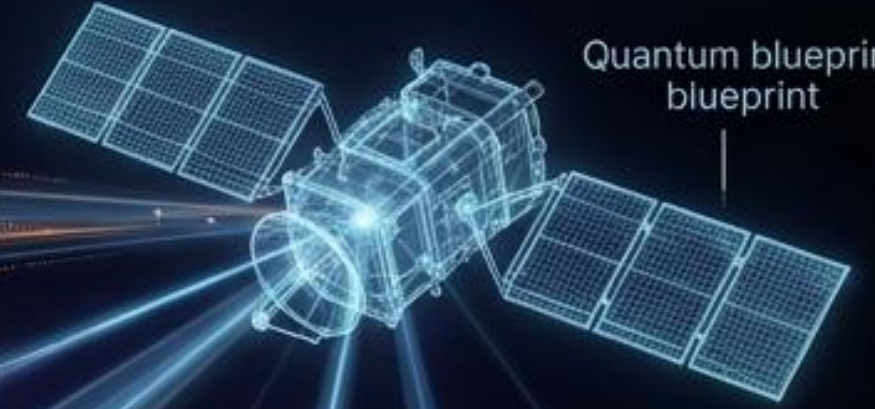
Virtualization and Quantum Mechanics Pushing SAGIN to the Absolute Limit

Space



Quantum Scheduling Matrix

Quantum blueprint blueprint



Air



Virtual Cybertwins



Ground



Holographic Network

Quantum-Optimized Virtualization



Global Access Scheduling via Quantum Computing

Coordinating thousands of Ground Stations, CubeSats, and HALE-UAVs surpasses classical computing limits.



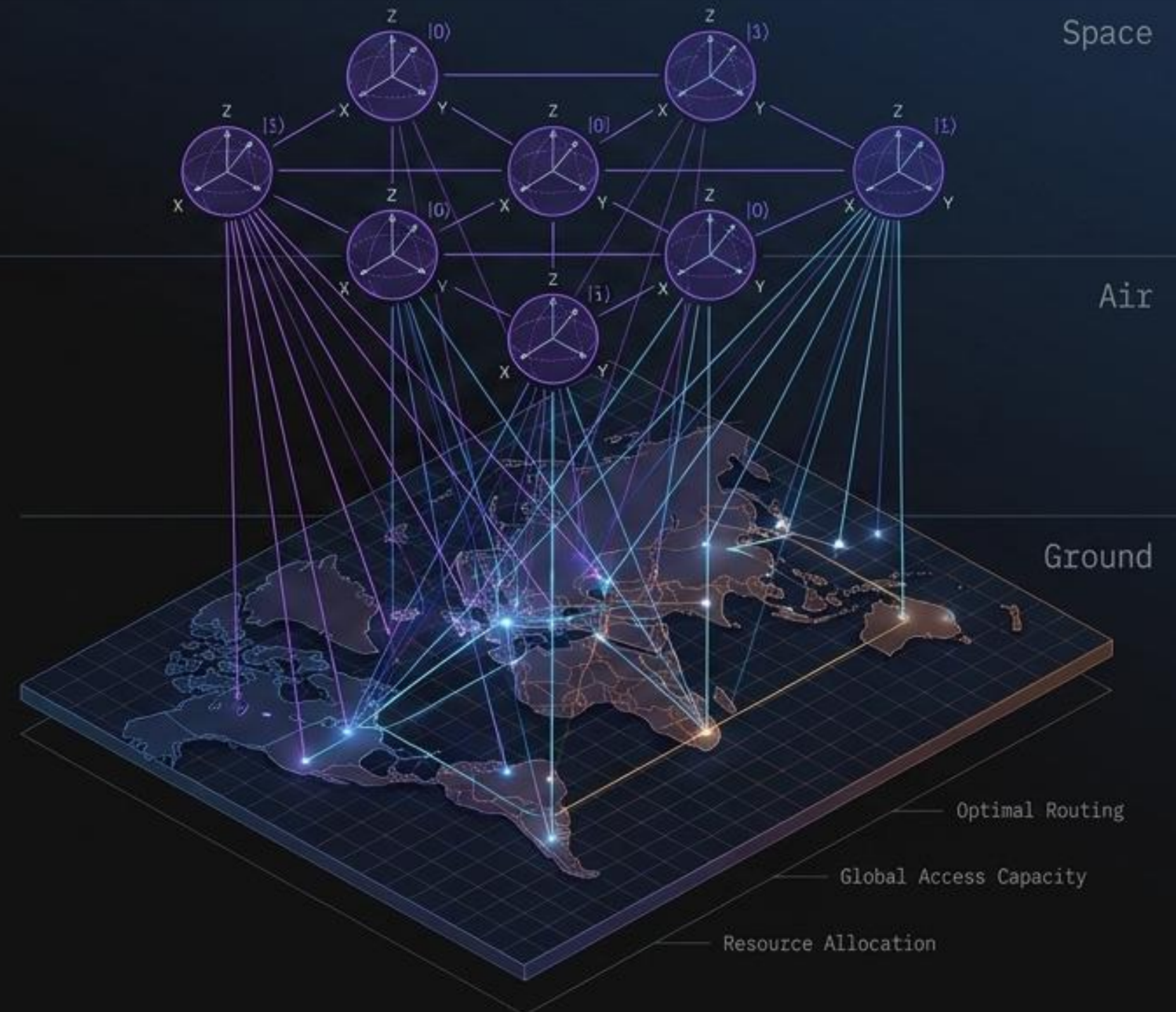
The Quantum Leap

Utilizing qubits as fundamental learning units ($|0\rangle$ and $|1\rangle$ superpositions).



Simultaneous Optimization

QNN maximizes residual energy, guarantees fairness, and optimizes global access capacity simultaneously across the entire Earth grid.



The Physics of Aerial Network Longevity

Sustaining High-Altitude Long-Endurance (HALE) UAVs requires precise AI management of aerodynamic drag.

Parasite Energy:

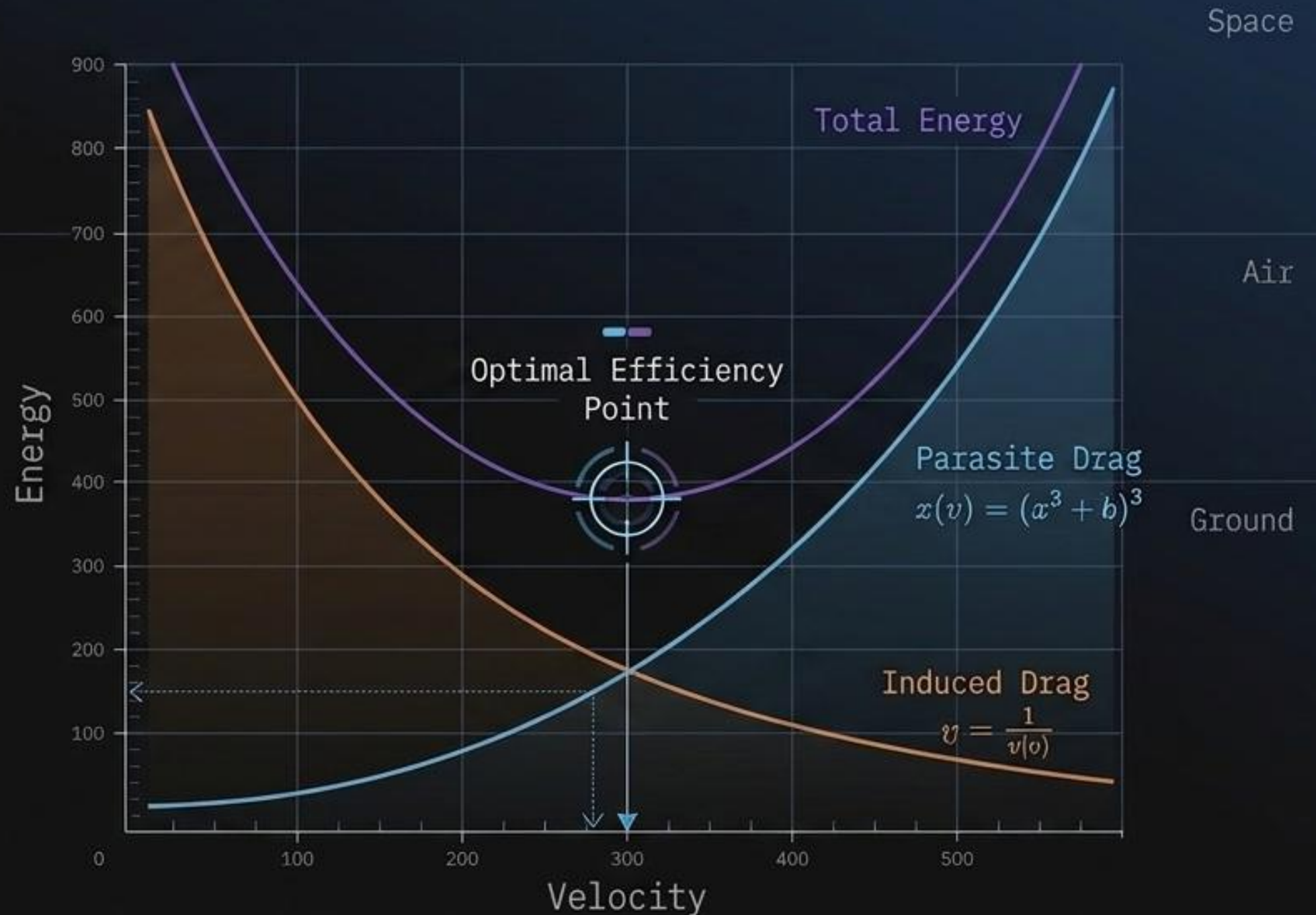
- Increases with the cube of velocity (skin friction and form drag).

Induced Energy:

- Inversely related to velocity (drag produced by generating lift).

Goal:

- Find the exact velocity to minimize total energy consumption, maximizing network availability.

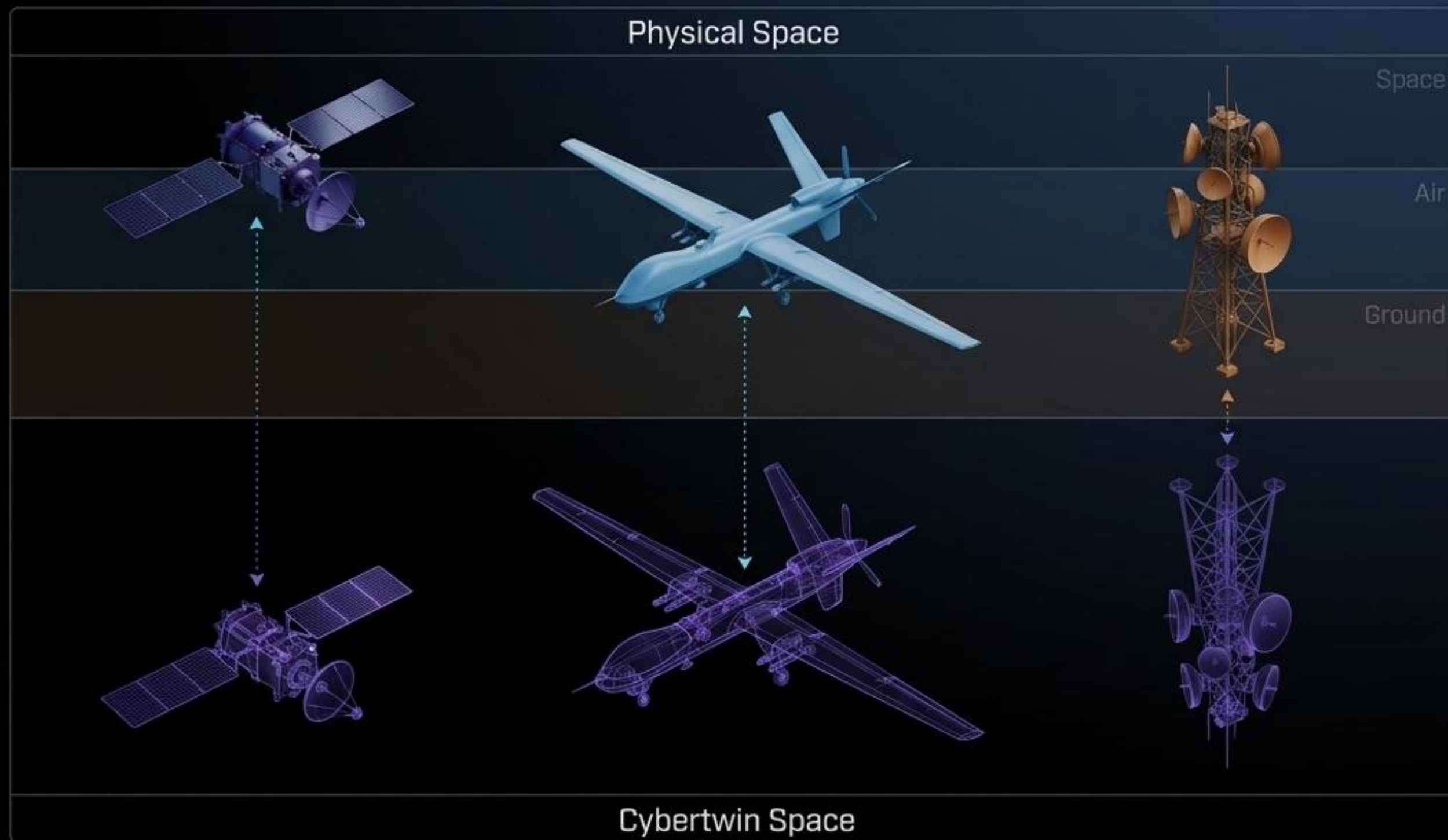


Cybertwins: Virtualizing the Physical Stratosphere

Every satellite, UAV, and base station possesses a digital twin, acting as an intelligent mobile agent.

Physical Space:
The hardware executing the final commands (LEOs, UAVs, Core Clouds).

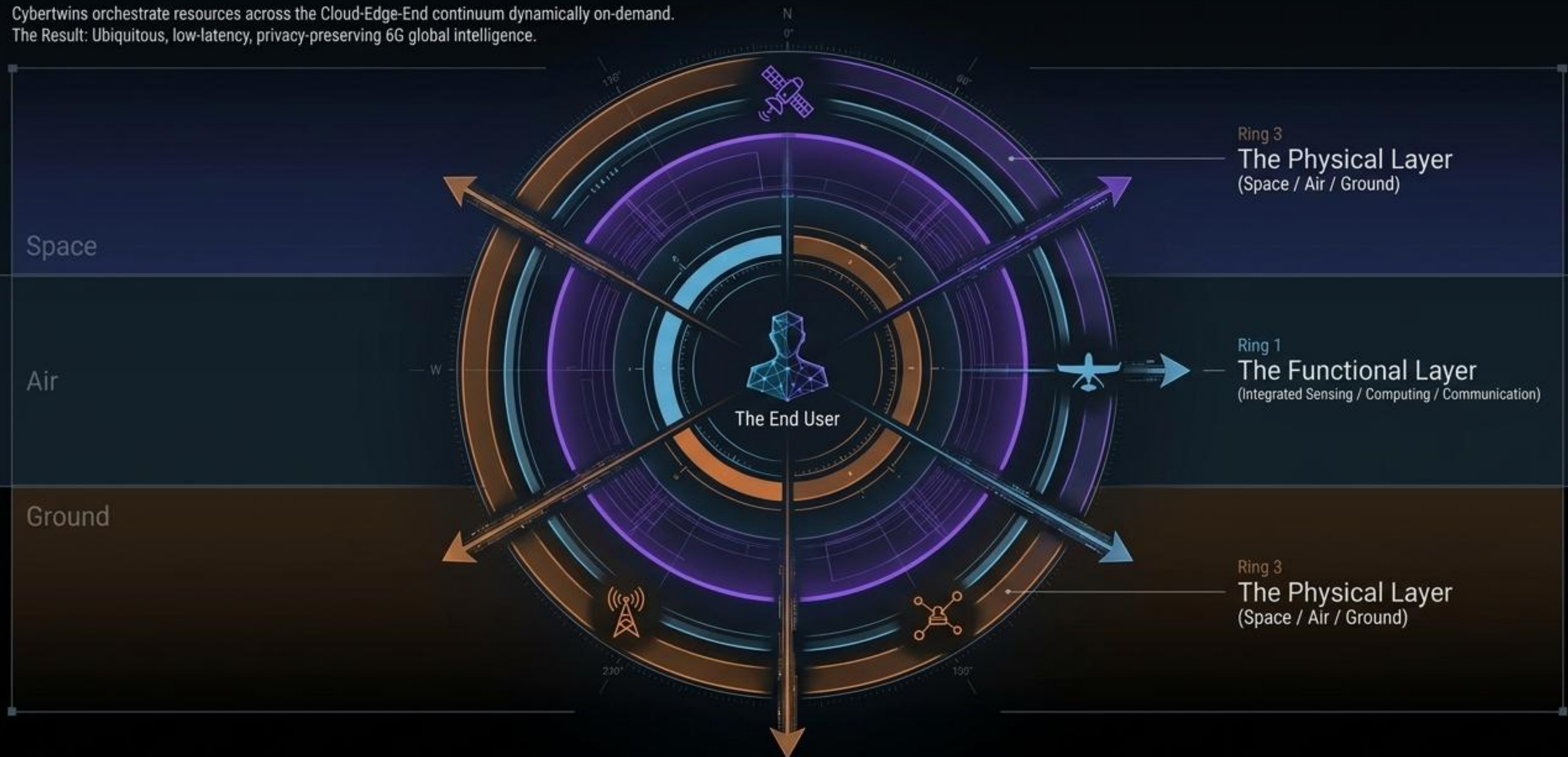
Cybertwin Space:
Digital reflections that perform data logging, federated model training, and decision-making virtually before physical execution.



Integrated Sensing, Communication, and Computing

The ultimate end-state of SAGIN: A multi-source heterogeneous network fully virtualized and integrated by AI.

Cybertwins orchestrate resources across the Cloud-Edge-End continuum dynamically on-demand.
The Result: Ubiquitous, low-latency, privacy-preserving 6G global intelligence.





Han et al. (2025) — THz Communication and Radar Sensing

My Paper Summaries: Intelligent Communication & Resource Management

1. An et al. (2026) – Electromagnetic Situation Awareness

- Uses AI, semantic analysis, and digital twins.
- Models spectrum conditions and interference.
- Improves awareness of the wireless environment.

2. Han et al. (2025) – THz Communication & Radar Sensing

- Uses Terahertz frequencies for ultra-fast communication.
- Enables both data transmission and environmental sensing.
- Challenges include path loss, mobility, and hardware limitations.

3. Huang et al. (2024) – Generative Semantic Communication

- Applies deep reinforcement learning.
- Optimizes semantic image transmission.
- Supports communication between satellites, UAVs, and ground users.

4. Zhao et al. (2025) – Energy-Efficient Semantic Communication

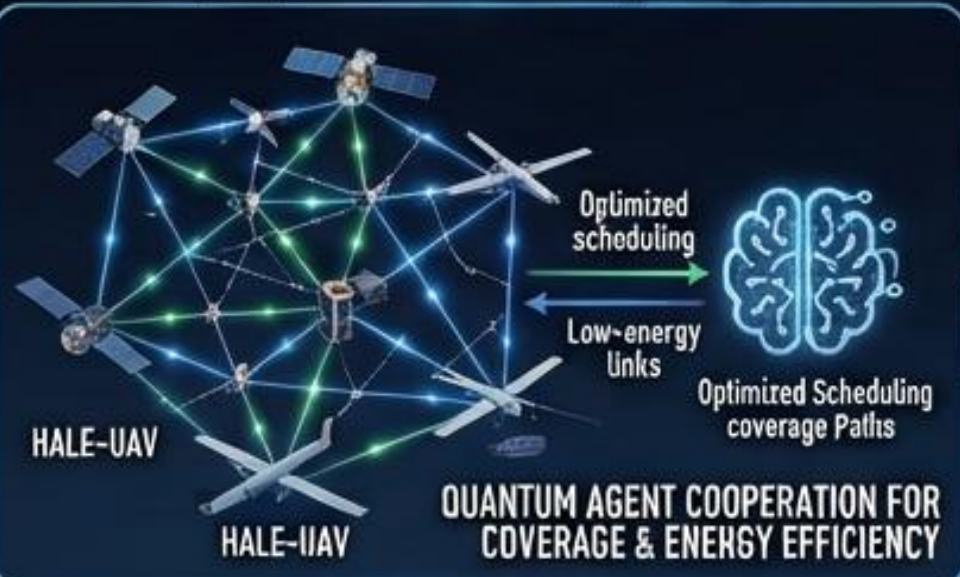
- Uses semantic compression to reduce data transmission.
- Optimizes UAV position, bandwidth, computing, and transmission power.
- Improves energy efficiency.

5. Mamillapalli et al. (2026) – RIS-Aided Interference Suppression

- Uses deep reinforcement learning for RIS beamforming.
- Reduces interference between different network layers.
- Improves spectrum efficiency and communication quality.

INTEGRATED AI, QUANTUM, BLOCKCHAIN, & CYBERTWIN ENABLED SPACE-AIR-GROUND INTEGRATED NETWORK (SAGIN)

Kim et al. (2024) — Quantum Multi-Agent Reinforcement Learning (Q-MARL)



Wu et al. (2024) — AI-Enabled Deployment Automation



Yin et al. (2022) — Cybertwin-Enabled SAGIN



Zhang (2025) — AI Model Collaboration

AI MODEL COLLABORATION & INFERENCE SHARING



Cheng et al. (2023) — Blockchain-Assisted Symbiotic Radio



Thanks for
listening

